Here $E(X^2) = \sum_{x=0}^{n} x^2 \binom{n}{x} p^x q^{n-x}$

$$\sum_{x=0}^{n} \{x(x-1)+x\} \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=0}^{n} \{x(x-1)\} \binom{n}{x} p^x q^{n-x} + \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{\epsilon=2}^{n} \{x(x-1)\} p^2 \left(\frac{n(n-1)}{x(x-1)} \right) \binom{n-2}{x-2} p^{x-2} q^{n-x} + \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

$$= n(n-1)\, p^2 \{ \sum_{x=2}^{n} \binom{n-2}{x-2} p^{x-2} q^{n-x} \} + np$$

$$= n(n-1)p^2(q+p)^{(n-2)} + np$$

$$= n(n-1)p^2 + np$$

$\therefore$ Variance $= E(X^2) - \{E(X)\}^2$

$$= n^2p^2 - np^2 + np - n^2p^2$$

$$= np(1-p) = npq$$

Hence, mean of the BD is np and the Variance is npq.

Note

Mean and variance in terms of raw moments and central moments are denoted as μ'_1, and μ_2 respectively.

Properties of Binomial distribution

1. Binomial distribution is symmetrical if $p = q = 0.5$. It is skew symmetric if $p \neq q$. It is positively skewed if $p < 0.5$ and it is negatively skewed if $p > 0.5$
2. For Binomial distribution, variance is less than mean

Variance $npq = (np)q < np$

Example 7.1

A and *B* play a game in which their chance of winning are in the ratio 3:2 Find *A*'s chance of winning atleast three games out of five games played.

Soltion:

Let '*p*' be the probability that '*A*' wins the game. Then we are given $n = 5$, $p = 3/5$,

$q = 1 - \frac{3}{5} = \frac{2}{5}$ (since $q = 1-p$)

Hence by binomial probability law, the probability that out of the 5 games played, *A* wins '*x*' games is given by

$$P(X=x) = p(x) = 5Cx \left(\frac{3}{5}\right)^x \left(\frac{2}{5}\right)^{5-x}$$

The required probability that '*A*' wins atleast three games is given by

$$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

$$= 5C3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2 + 5C4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^1 + 5C5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^0$$

$= 0.6826$

Example 7.2

A fair coin is tossed 6 times. Find the probability that exactly 2 heads occurs.

Solution :

Let *X* be a random variable follows binomial distribution with probability value $p = 1/2$ and $q = 1/2$

Probability that exactly 2 heads occur are as follows

$$P(X=2) = \binom{6}{x} p^x q^{n-x}$$

$$= \binom{6}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{6-2}$$

$$= \frac{15}{64}$$

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